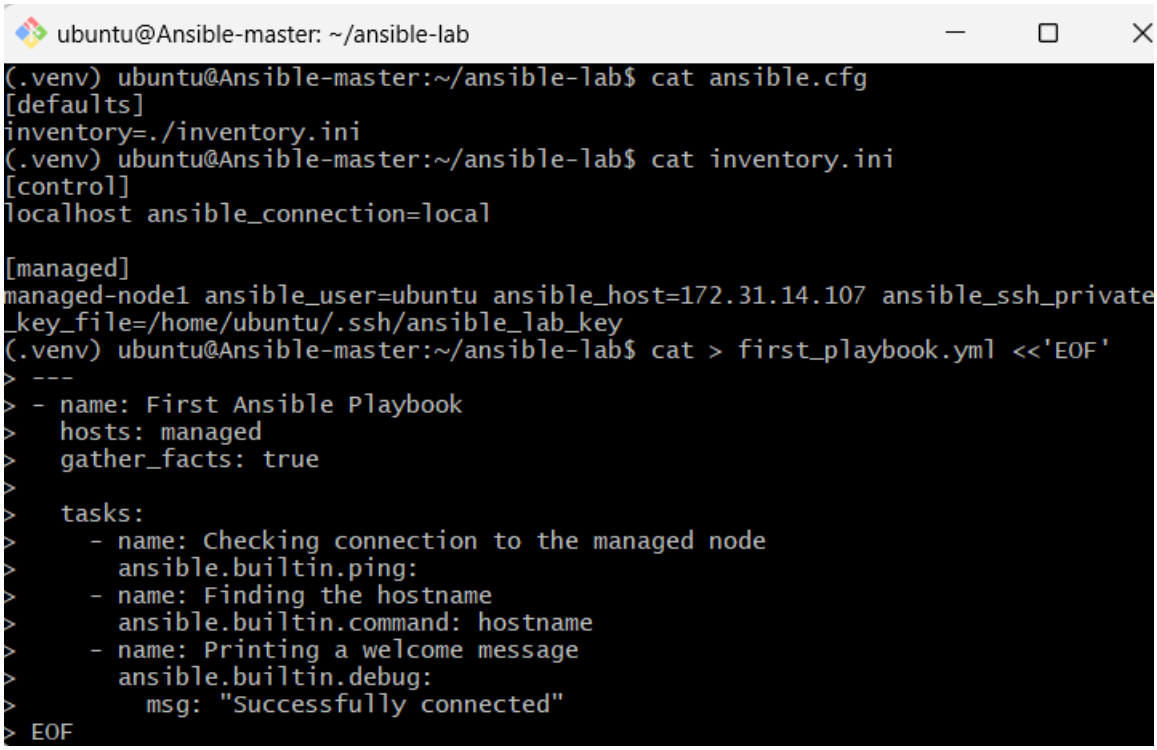
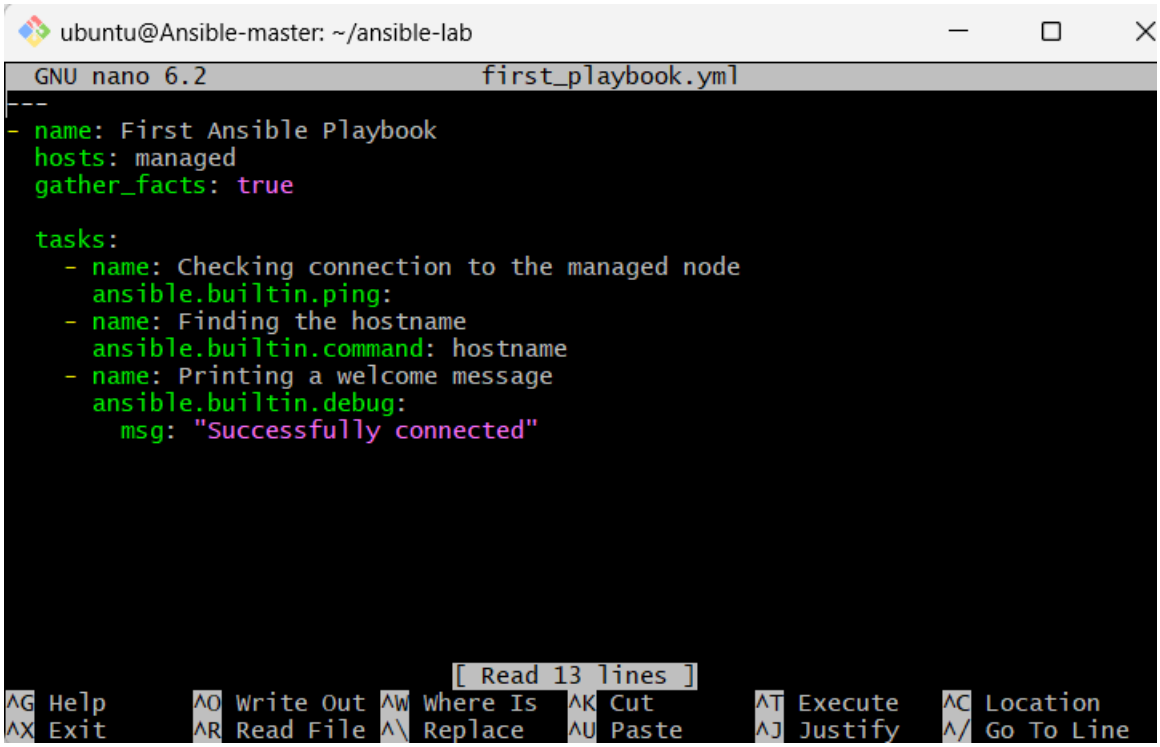
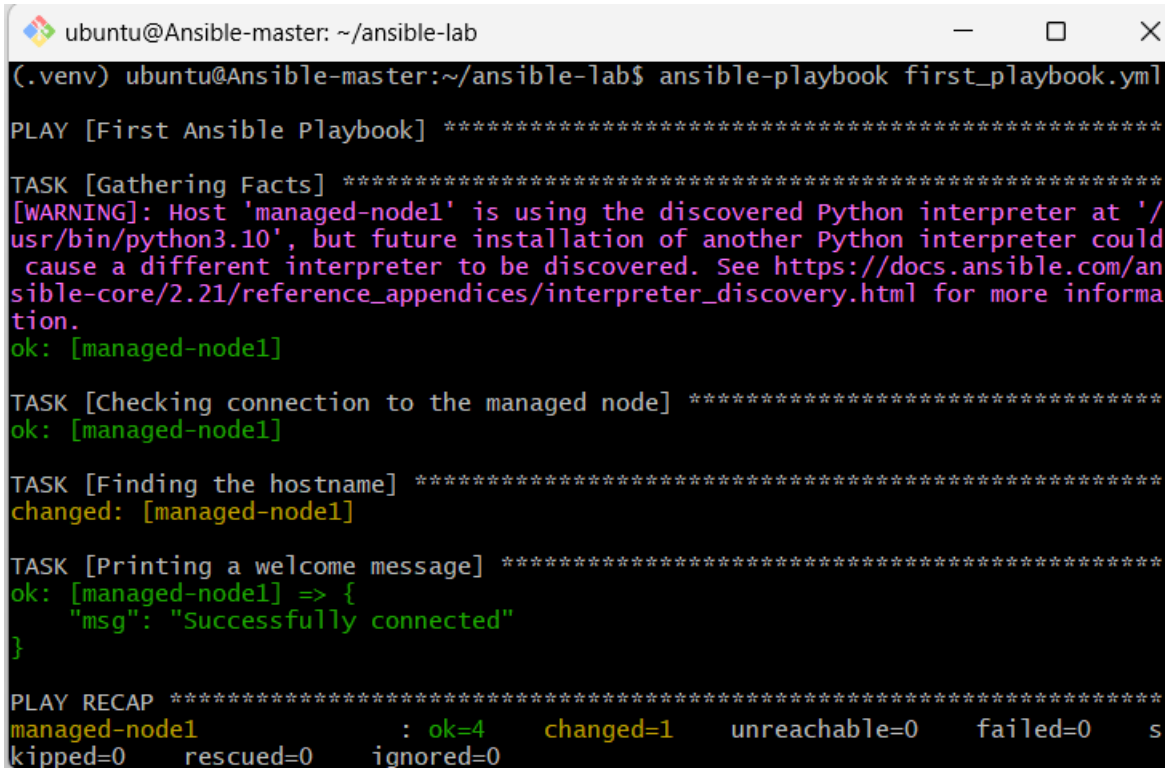
Ansible Mastery Part-2:

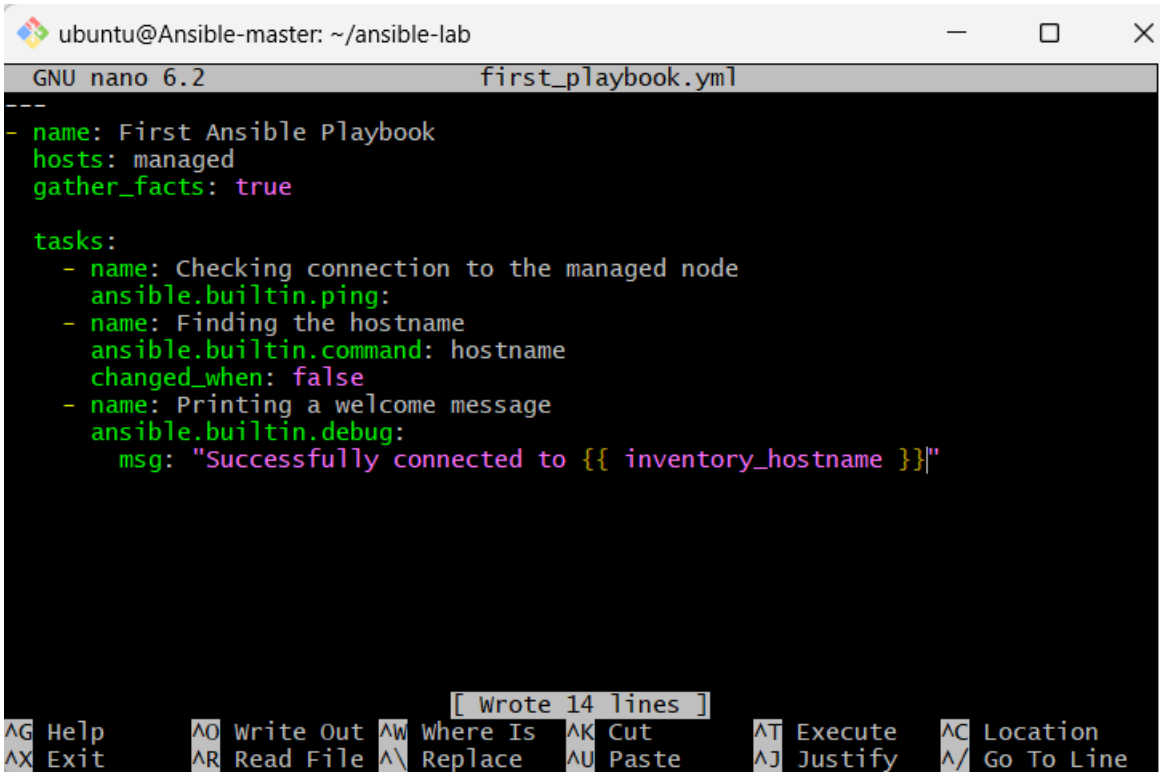
Ansible Playbook Fundamentals: Idempotent Configuration Management, Nginx Service Deployment, Handler Execution, File and User Management, and Error Handling



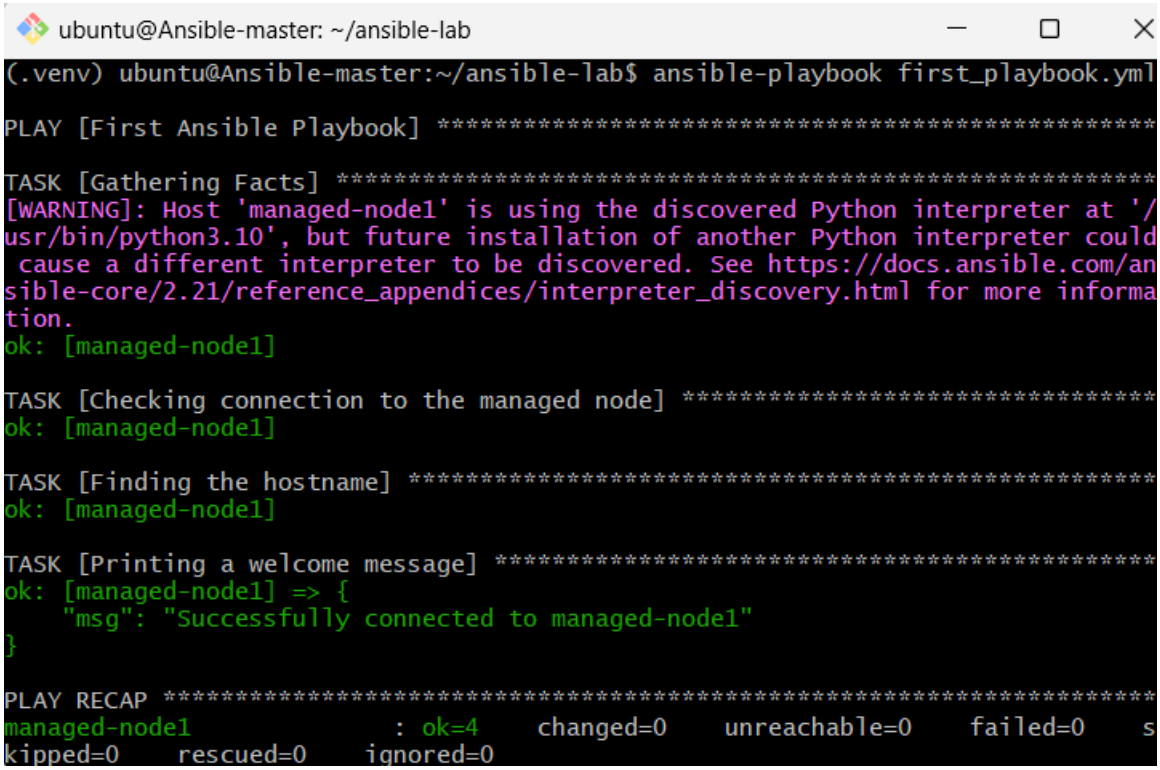

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 first_playbook.yml

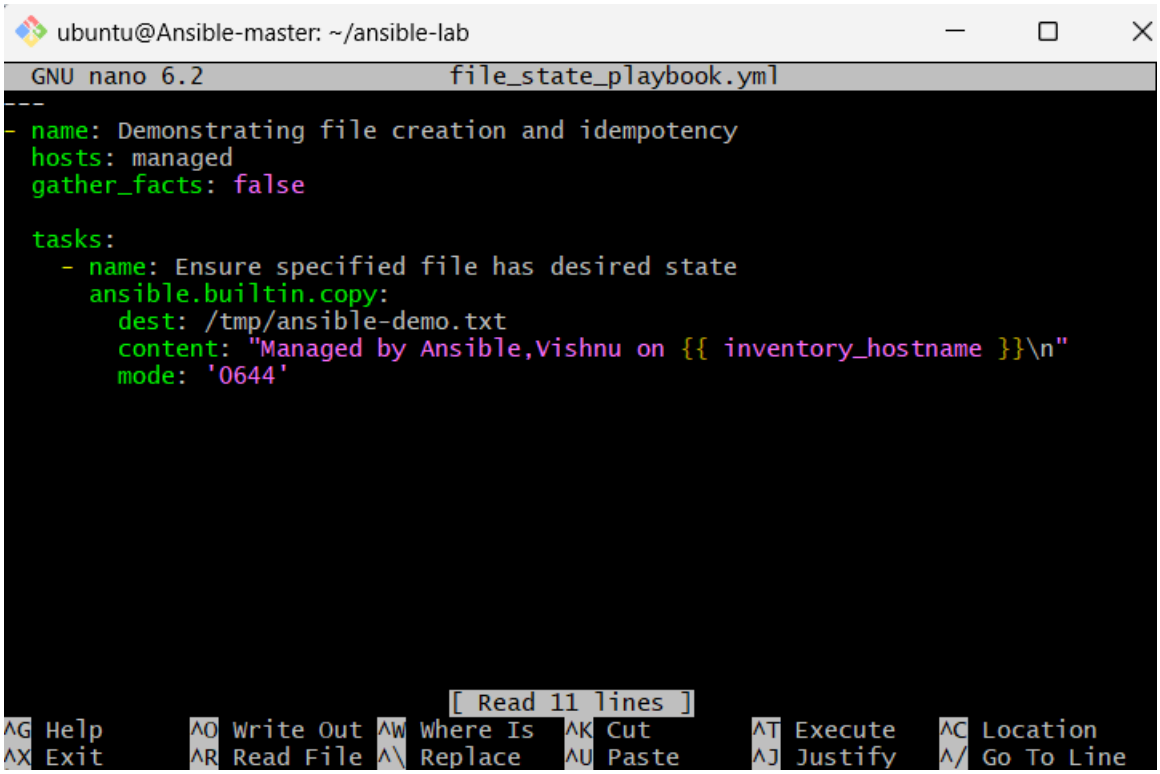
- name: First Ansible Playbook
hosts: managed
gather_facts: true
tasks:
- name: Checking connection to the managed node
ansible.builtin.ping:
- name: Finding the hostname
ansible.builtin.command: hostname
- name: Printing a welcome message
ansible.builtin.debug:
msg: "Successfully connected"
[Read 13 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook first_playbook.yml
PLAY [First Ansible Playbook]
TASK [Gathering Facts]
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [managed-node1]
TASK [Checking connection to the managed node]
ok: [managed-node1]
TASK [Finding the hostname]
changed: [managed-node1]
TASK [Printing a welcome message]
ok: [managed-node1] => {
"msg": "Successfully connected"
}
PLAY RECAP
managed-node1 : ok=4 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

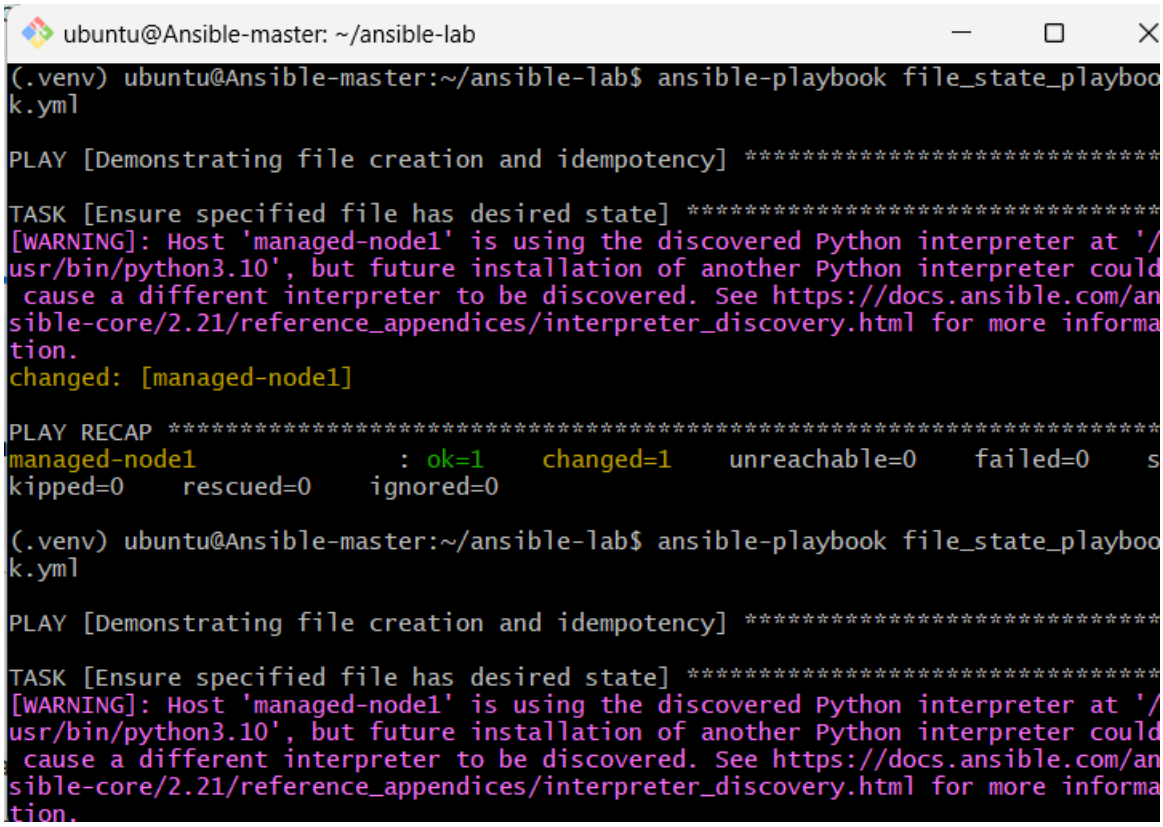


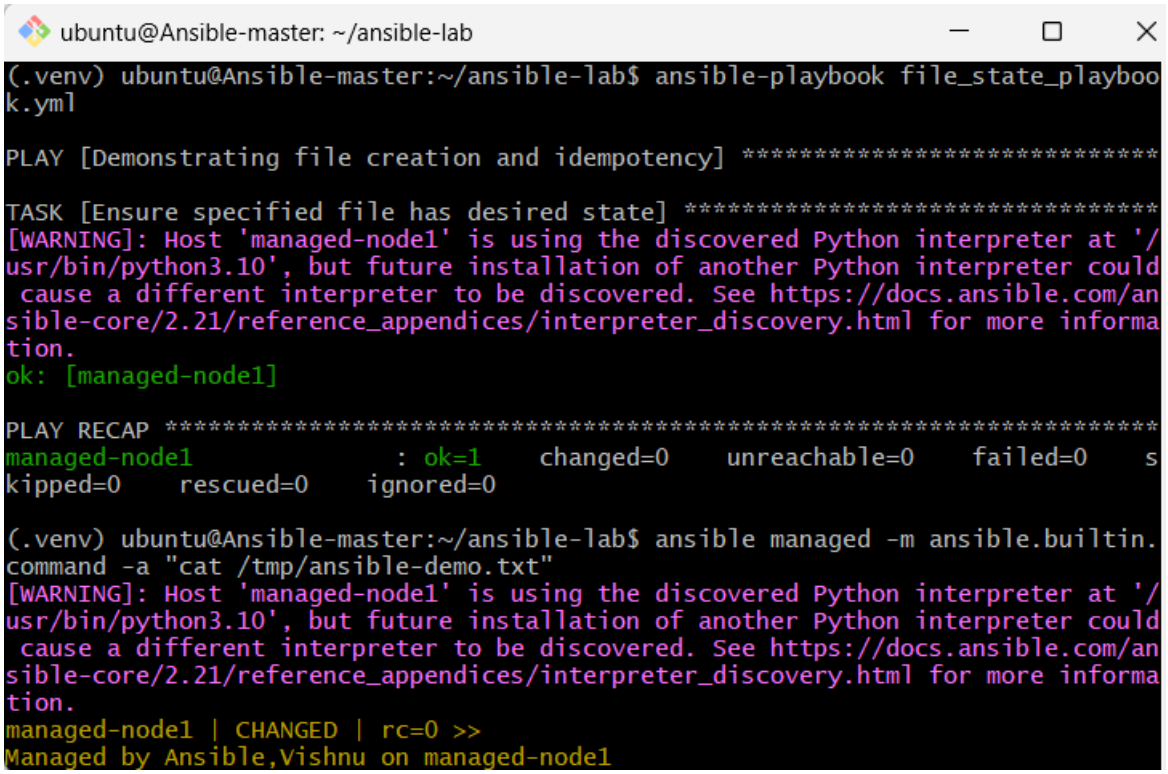
Note: Jinja2 dynamically replaces {{ inventory_hostname }} with the current host's inventory name at runtime.




ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 file_state_playbook.yml

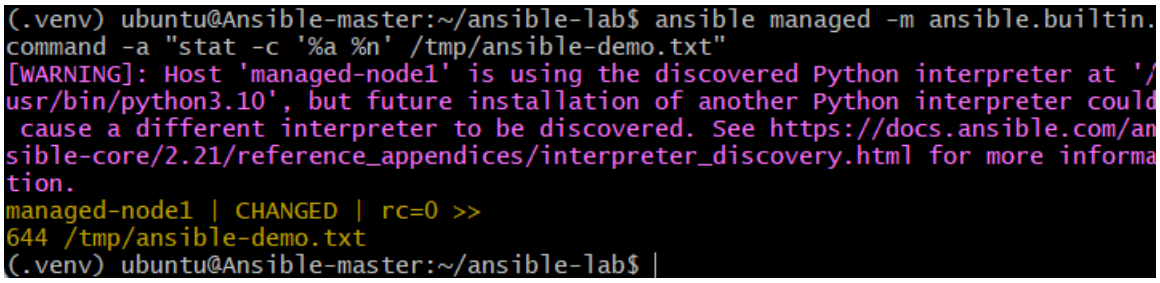
- name: Demonstrating file creation and idempotency
hosts: managed
gather_facts: false
tasks:
- name: Ensure specified file has desired state
ansible.builtin.copy:
dest: /tmp/ansible-demo.txt
content: "Managed by Ansible,Vishnu on {{ inventory_hostname }}\n"
mode: '0644'
[Read 11 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

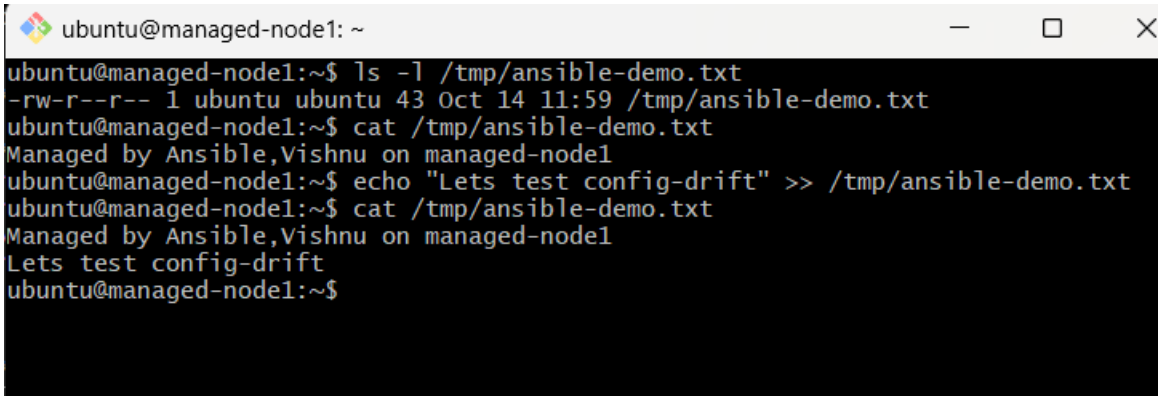

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook file_state_playboo
k.yml
PLAY [Demonstrating file creation and idempotency] ****************************
TASK [Ensure specified file has desired state] ********************************
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
changed: [managed-node1]
PLAY RECAP **
managed-node1 : ok=1 changed=1 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook file_state_playboo
k.yml
PLAY [Demonstrating file creation and idempotency] ****************************
TASK [Ensure specified file has desired state] ********************************
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.

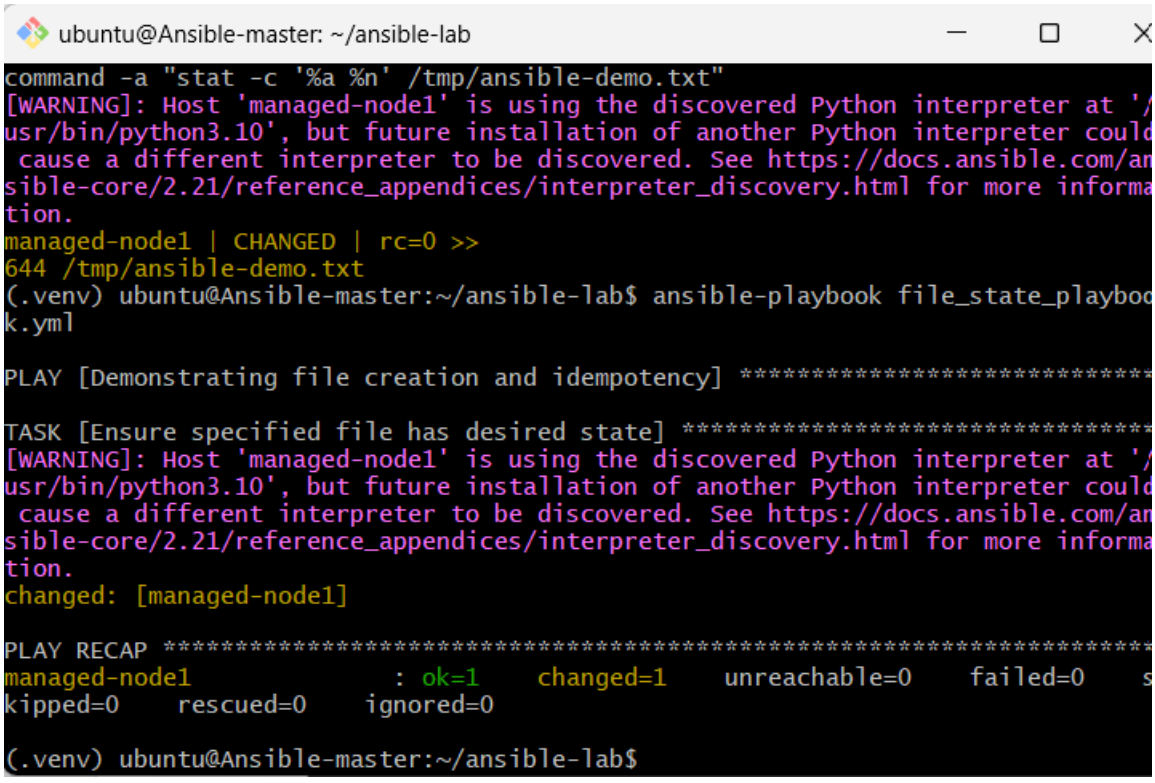


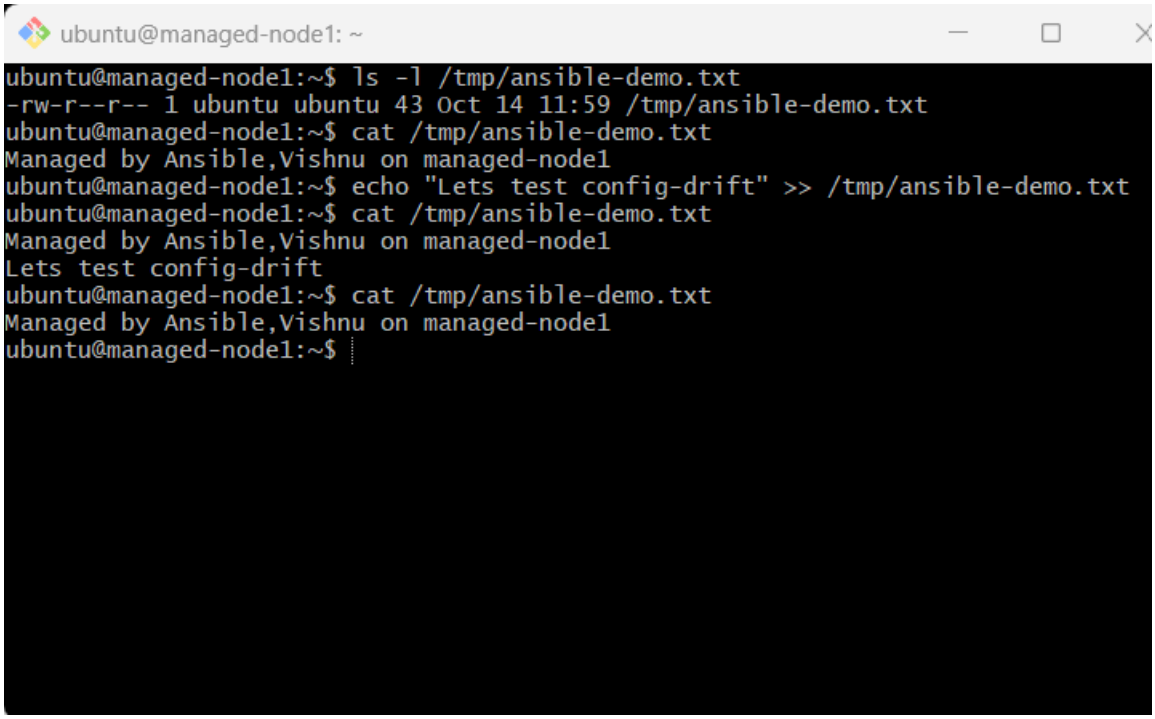
Note: Ansible idempotency: the first run creates or updates the file; subsequent runs report changed=0 when the desired state is already satisfied.

Configuration drift detection and desired-state enforcement



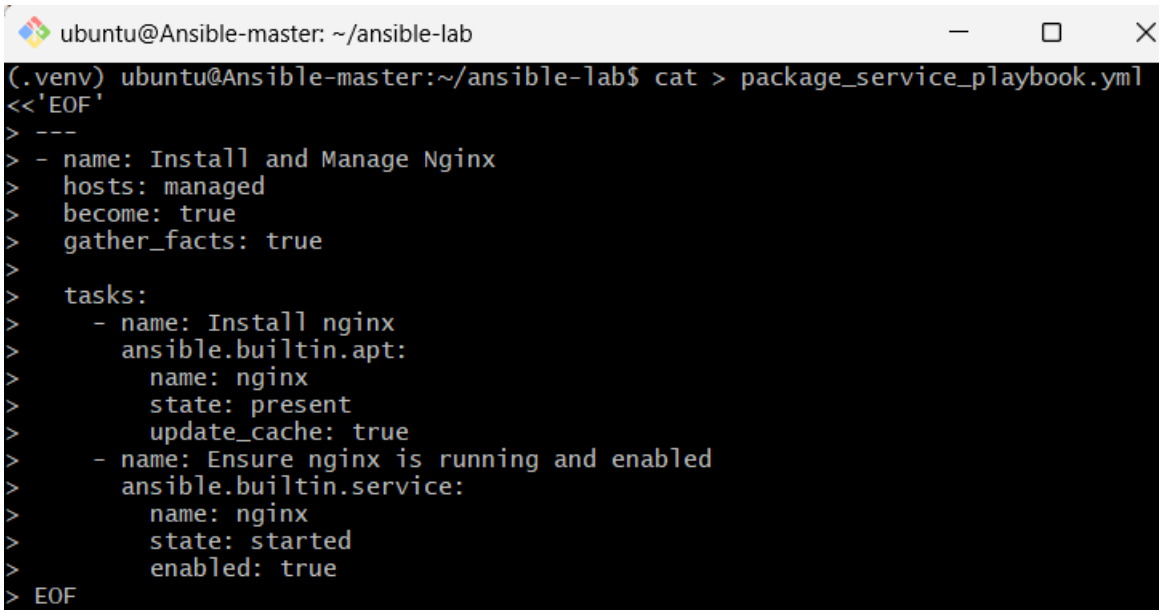

ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ ls -l /tmp/ansible-demo.txt
-rw-r--r-- 1 ubuntu ubuntu 43 Oct 14 11:59 /tmp/ansible-demo.txt
ubuntu@managed-node1:~$ cat /tmp/ansible-demo.txt
Managed by Ansible,Vishnu on managed-node1
ubuntu@managed-node1:~$ echo "Lets test config-drift" >> /tmp/ansible-demo.txt
ubuntu@managed-node1:~$ cat /tmp/ansible-demo.txt
Managed by Ansible,Vishnu on managed-node1
Lets test config-drift
ubuntu@managed-node1:~$

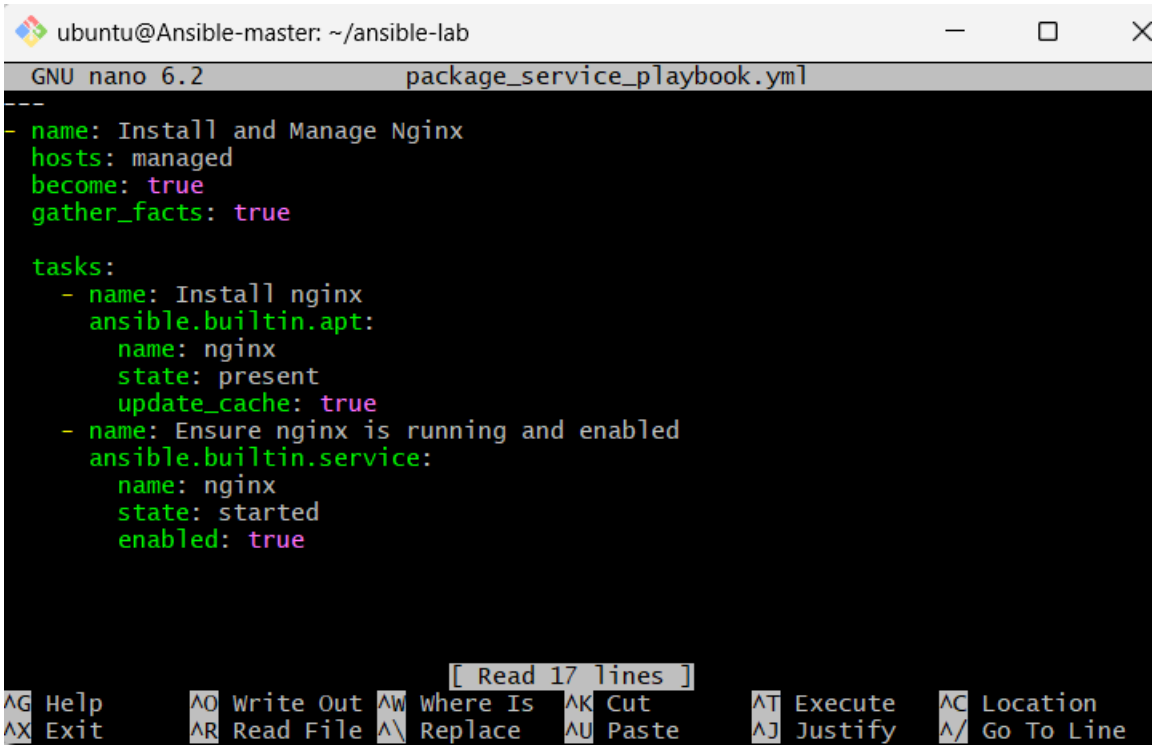

ubuntu@Ansible-master: ~/ansible-lab
command -a "stat -c '%a %n' /tmp/ansible-demo.txt"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
managed-node1 | CHANGED | rc=0 >>
644 /tmp/ansible-demo.txt
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook file_state_playboo
k.yml
PLAY [Demonstrating file creation and idempotency]
TASK [Ensure specified file has desired state]
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
changed: [managed-node1]
PLAY RECAP
managed-node1 : ok=1 changed=1 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$



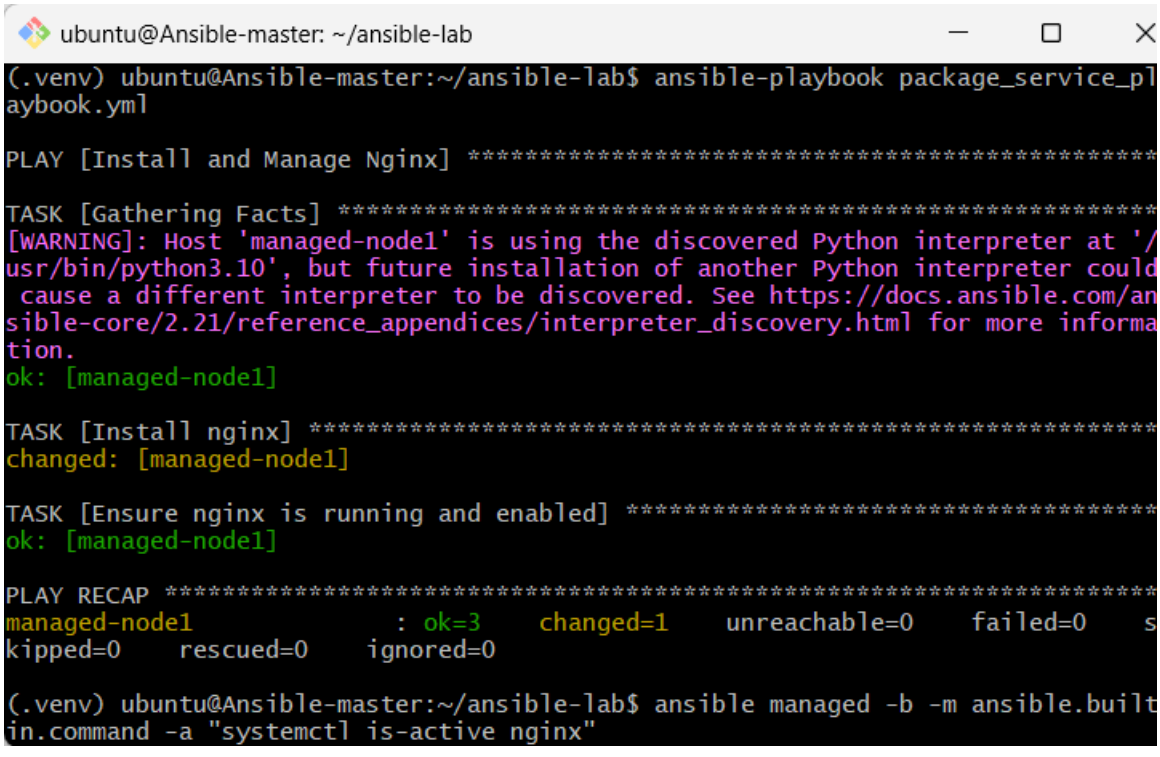
Desired state achieved

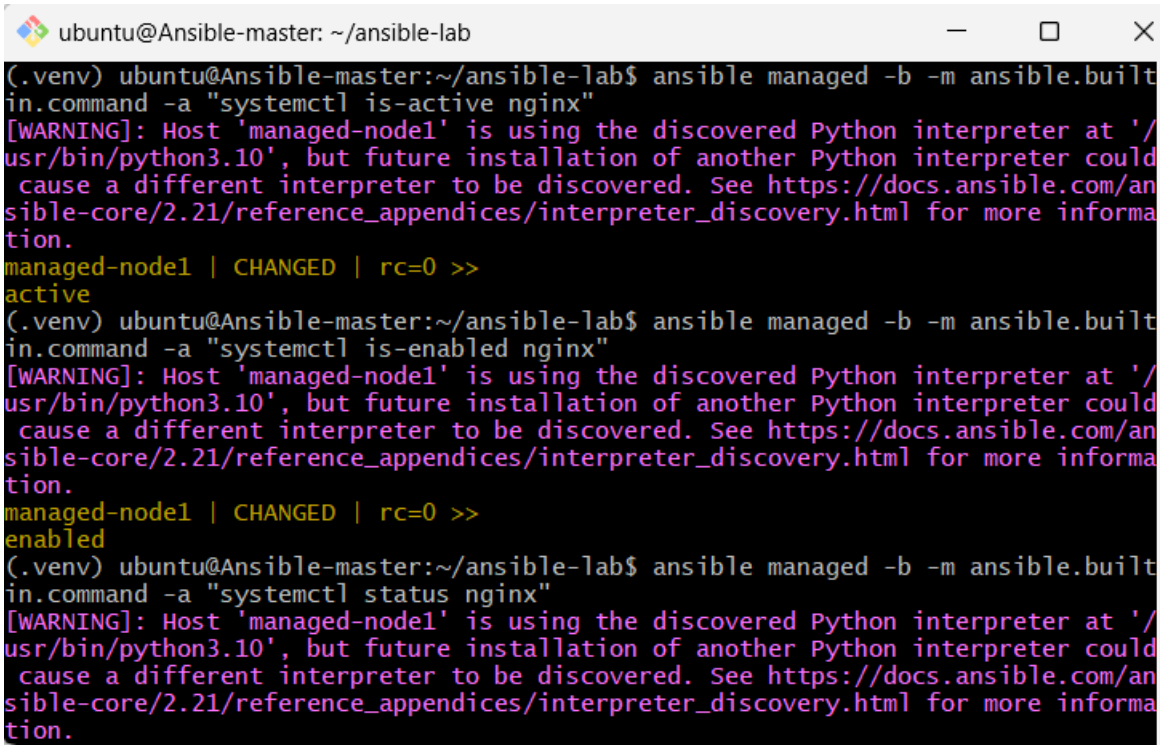
Nginx Installation and Service Management in managed-node through Ansible

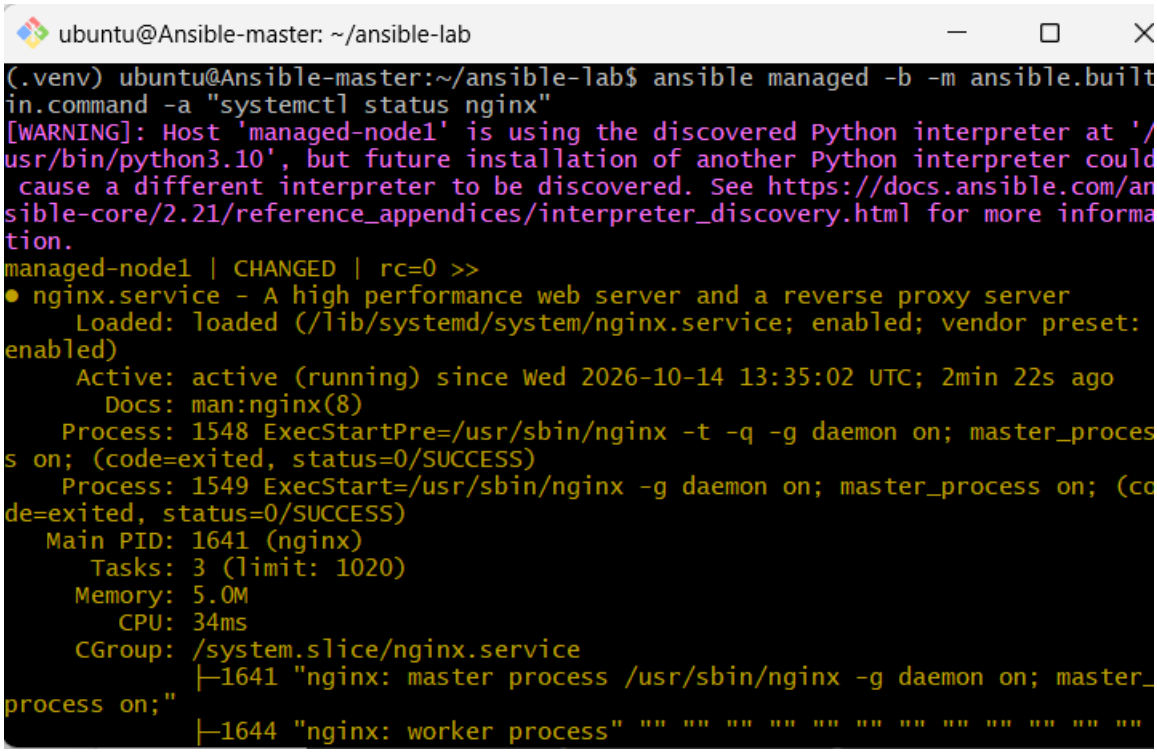


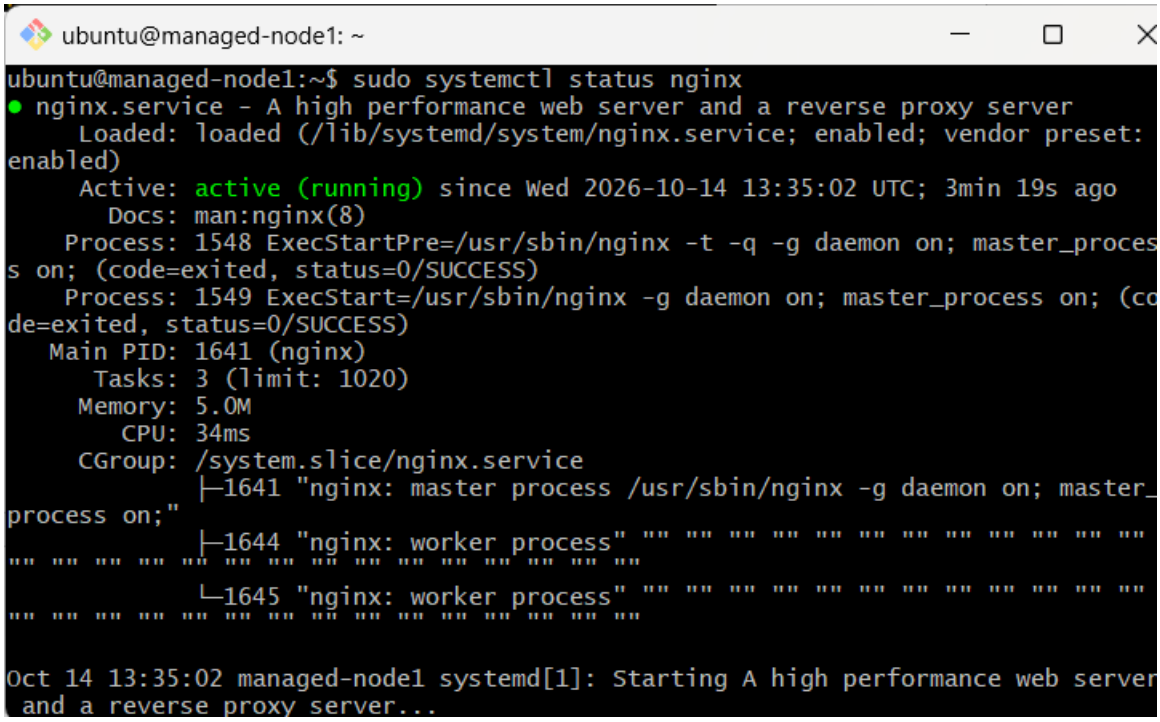

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 package_service_playbook.yml

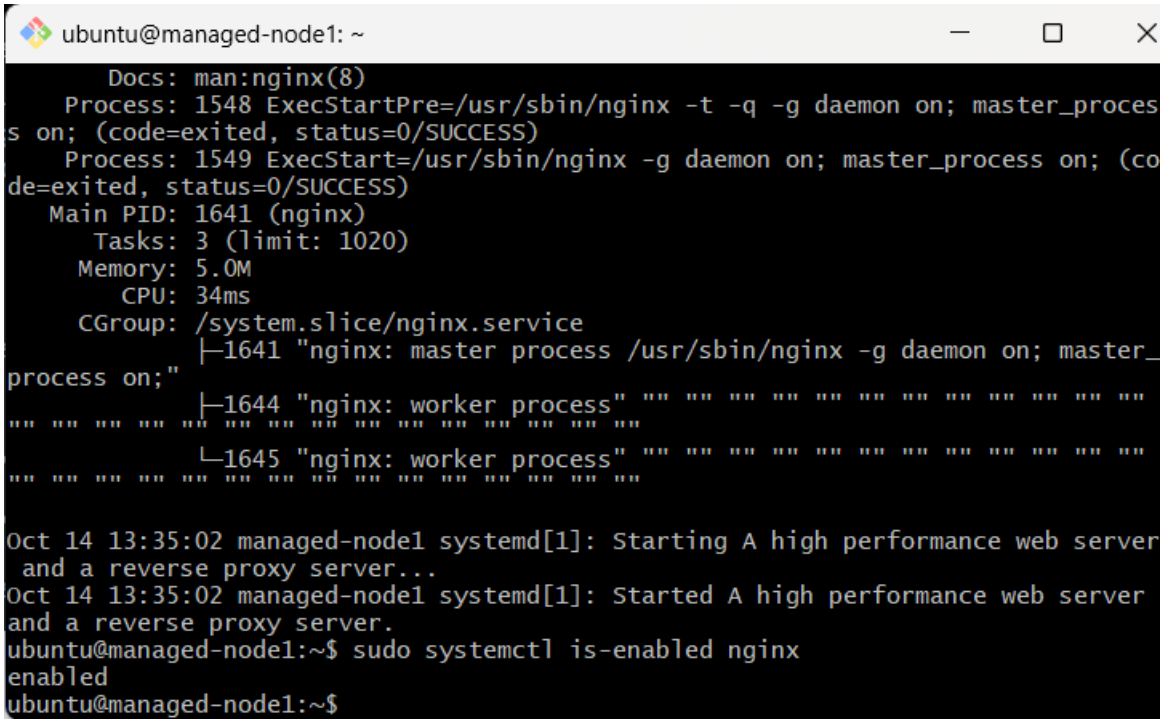
- name: Install and Manage Nginx
hosts: managed
become: true
gather_facts: true
tasks:
- name: Install nginx
ansible.builtin.apt:
name: nginx
state: present
update_cache: true
- name: Ensure nginx is running and enabled
ansible.builtin.service:
name: nginx
state: started
enabled: true
[Read 17 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

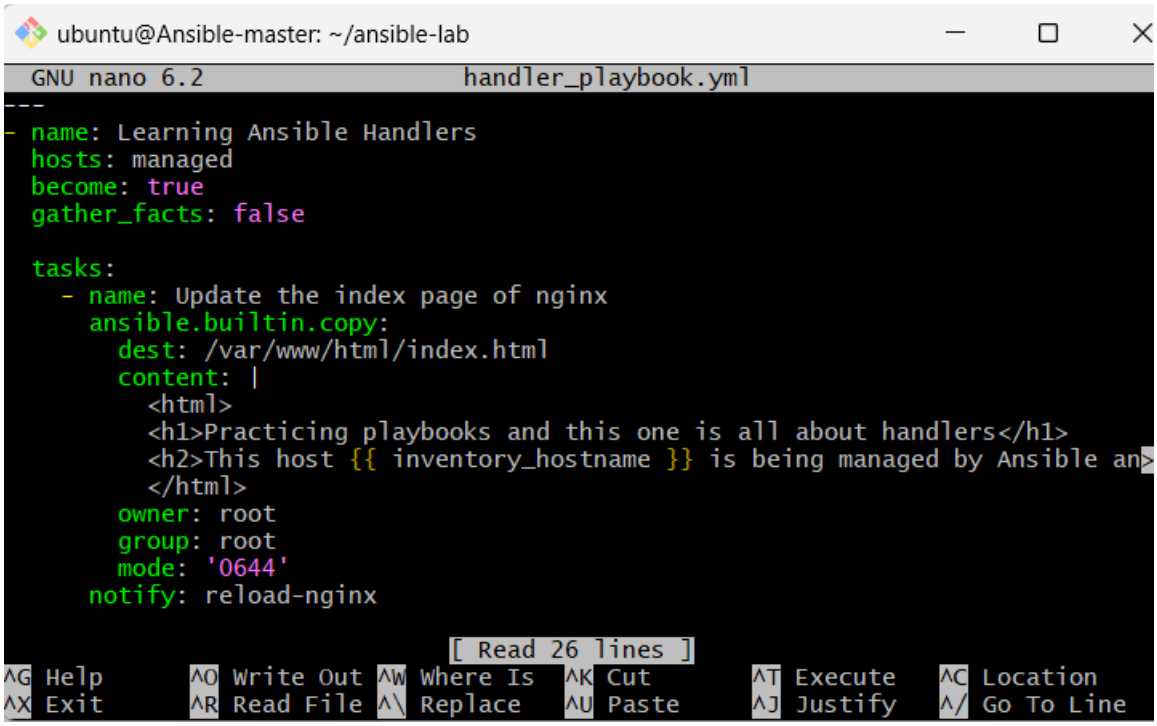

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook package_service_playbook.yml
PLAY [Install and Manage Nginx] ****
TASK [Gathering Facts] ****
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [managed-node1]
TASK [Install nginx] ****
changed: [managed-node1]
TASK [Ensure nginx is running and enabled] ****
ok: [managed-node1]
PLAY RECAP ****
managed-node1 : ok=3 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.builtin.command -a "systemctl is-active nginx"


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.built
in.command -a "systemctl is-active nginx"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
 cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
managed-node1 | CHANGED | rc=0 >>
active
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.built
in.command -a "systemctl is-enabled nginx"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
 cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
managed-node1 | CHANGED | rc=0 >>
enabled
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.built
in.command -a "systemctl status nginx"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
 cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.

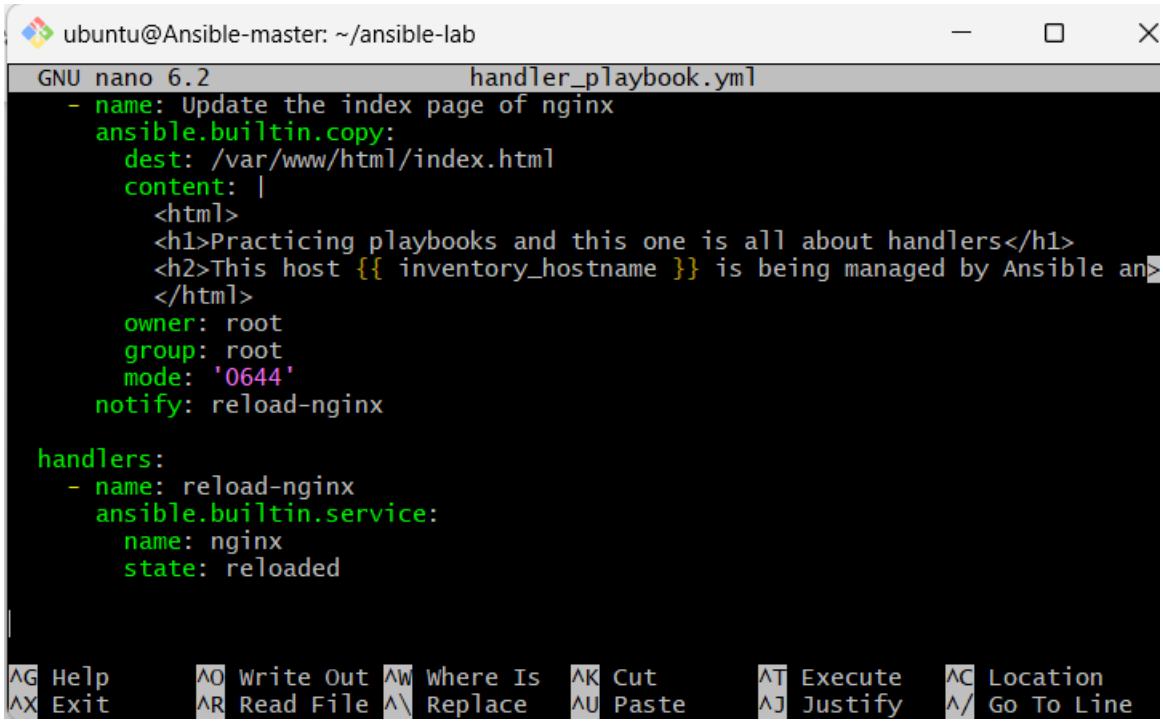

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.built
in.command -a "systemctl status nginx"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
 cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
managed-node1 | CHANGED | rc=0 >>
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
enabled)
     Active: active (running) since Wed 2026-10-14 13:35:02 UTC; 2min 22s ago
       Docs: man:nginx(8)
    Process: 1548 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proces
s on; (code=exited, status=0/SUCCESS)
    Process: 1549 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (co
de=exited, status=0/SUCCESS)
   Main PID: 1641 (nginx)
      Tasks: 3 (limit: 1020)
     Memory: 5.0M
        CPU: 34ms
     CGroup: /system.slice/nginx.service
             ├─1641 "nginx: master process /usr/sbin/nginx -g daemon on; master_
process on;"
             ├─1644 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" ""

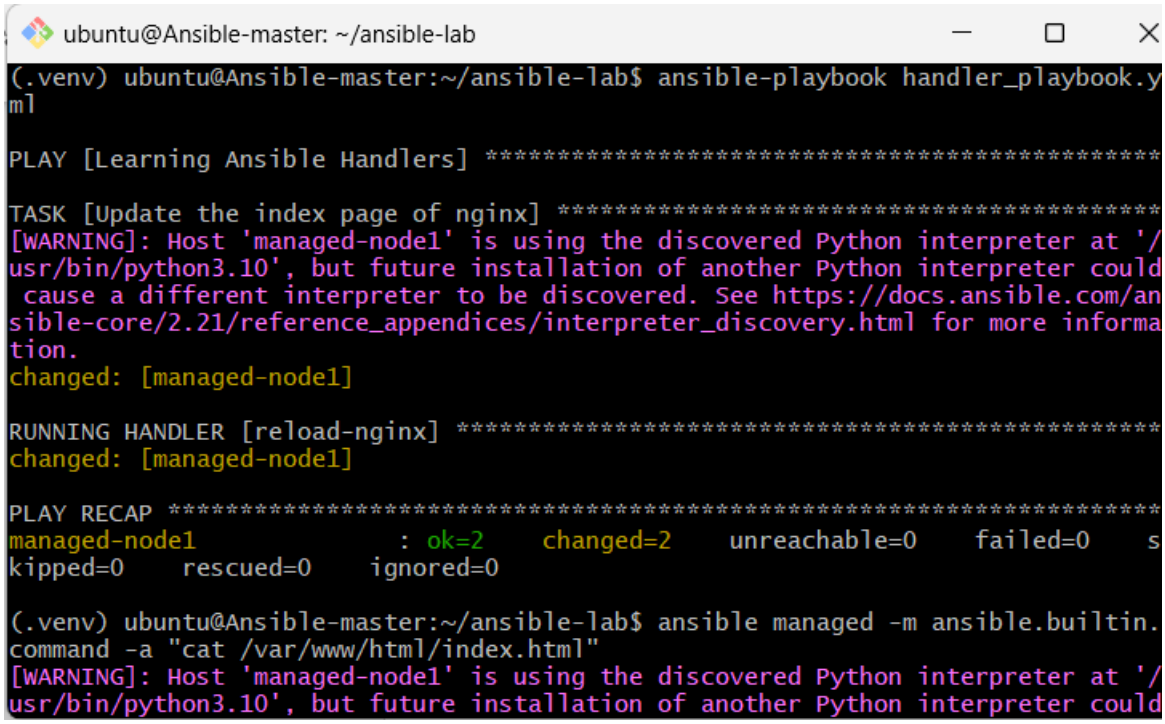

ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
enabled)
Active: active (running) since Wed 2026-10-14 13:35:02 UTC; 3min 19s ago
Docs: man:nginx(8)
Process: 1548 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proces
s on; (code=exited, status=0/SUCCESS)
Process: 1549 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (co
de=exited, status=0/SUCCESS)
Main PID: 1641 (nginx)
Tasks: 3 (limit: 1020)
Memory: 5.0M
CPU: 34ms
CGroup: /system.slice/nginx.service
├─1641 "nginx: master process /usr/sbin/nginx -g daemon on; master_
process on;"
├─1644 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "
"" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
└─1645 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "
"" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
Oct 14 13:35:02 managed-node1 systemd[1]: Starting A high performance web server
and a reverse proxy server...


ubuntu@managed-node1: ~
Docs: man:nginx(8)
Process: 1548 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proces
s on; (code=exited, status=0/SUCCESS)
Process: 1549 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (co
de=exited, status=0/SUCCESS)
Main PID: 1641 (nginx)
Tasks: 3 (limit: 1020)
Memory: 5.0M
CPU: 34ms
CGroup: /system.slice/nginx.service
├─1641 "nginx: master process /usr/sbin/nginx -g daemon on; master_
process on;"
├─1644 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "
"" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
└─1645 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "
"" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
Oct 14 13:35:02 managed-node1 systemd[1]: Starting A high performance web server
and a reverse proxy server...
Oct 14 13:35:02 managed-node1 systemd[1]: Started A high performance web server
and a reverse proxy server.
ubuntu@managed-node1:~$ sudo systemctl is-enabled nginx
enabled
ubuntu@managed-node1:~$

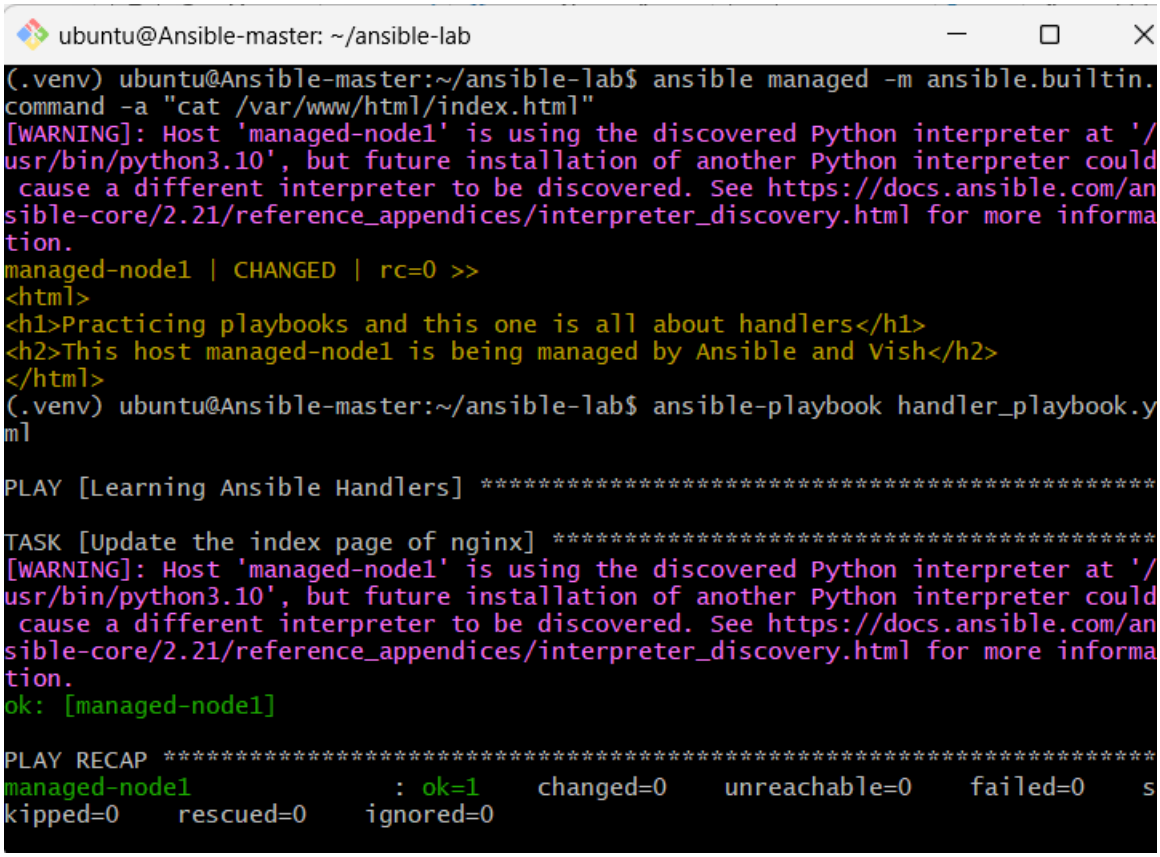

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 handler_playbook.yml

- name: Learning Ansible Handlers
hosts: managed
become: true
gather_facts: false
tasks:
- name: Update the index page of nginx
ansible.builtin.copy:
dest: /var/www/html/index.html
content: |
<html>
<h1>Practicing playbooks and this one is all about handlers</h1>
<h2>This host {{ inventory_hostname }} is being managed by Ansible an
</html>
owner: root
group: root
mode: '0644'
notify: reload-nginx
[Read 26 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 handler_playbook.yml
- name: Update the index page of nginx
ansible.builtin.copy:
dest: /var/www/html/index.html
content: |
<html>
<h1>Practicing playbooks and this one is all about handlers</h1>
<h2>This host {{ inventory_hostname }} is being managed by Ansible an
</html>
owner: root
group: root
mode: '0644'
notify: reload-nginx
handlers:
- name: reload-nginx
ansible.builtin.service:
name: nginx
state: reloaded
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line



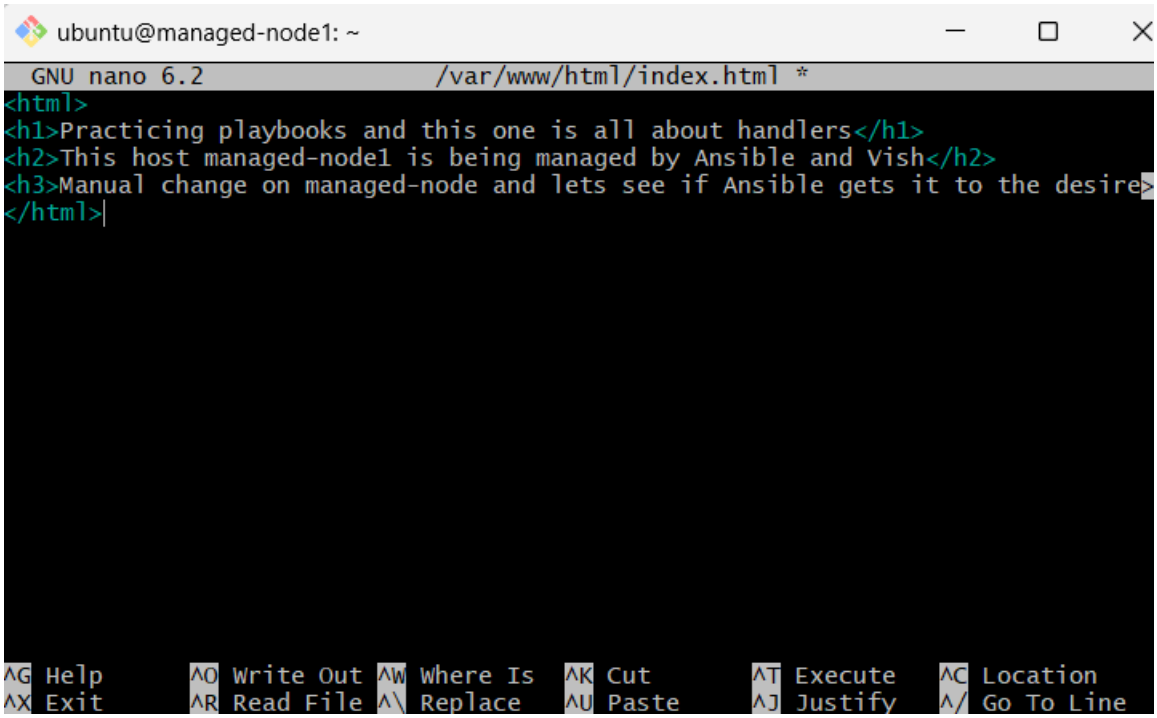
The handler ran successfully and reloaded the Nginx service after the change





Nginx served it from the managed-node

Manual Configuration drift and handler execution






```
ubuntu@Ansible-master: ~/ansible-lab
managed-node1              : ok=1    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

(.venv) ubuntu@Ansible-master:~/ansible-lab$ nano handler_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook handler_playbook.y
ml

PLAY [Learning Ansible Handlers] ***********************************************

TASK [Update the index page of nginx] ******************************************
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/
usr/bin/python3.10', but future installation of another Python interpreter could
 cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.21/reference_appendices/interpreter_discovery.html for more informa
tion.
changed: [managed-node1]

RUNNING HANDLER [reload-nginx] *************************************************
changed: [managed-node1]

PLAY RECAP *********************************************************************
managed-node1              : ok=2    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

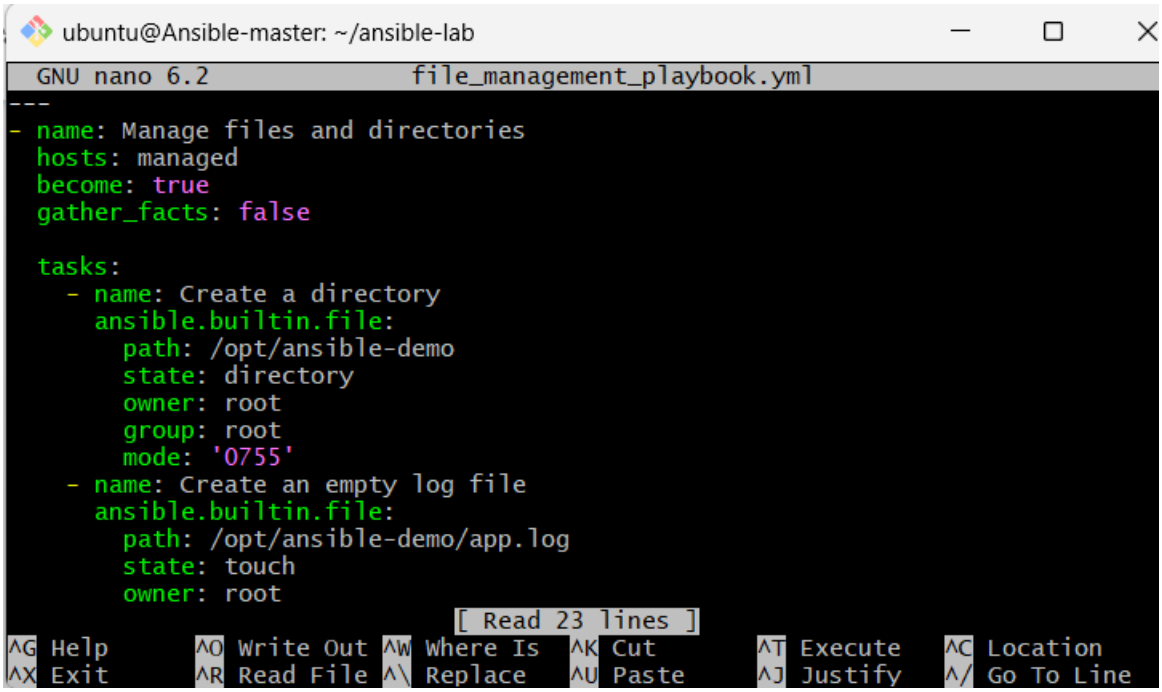
(.venv) ubuntu@Ansible-master:~/ansible-lab$
```

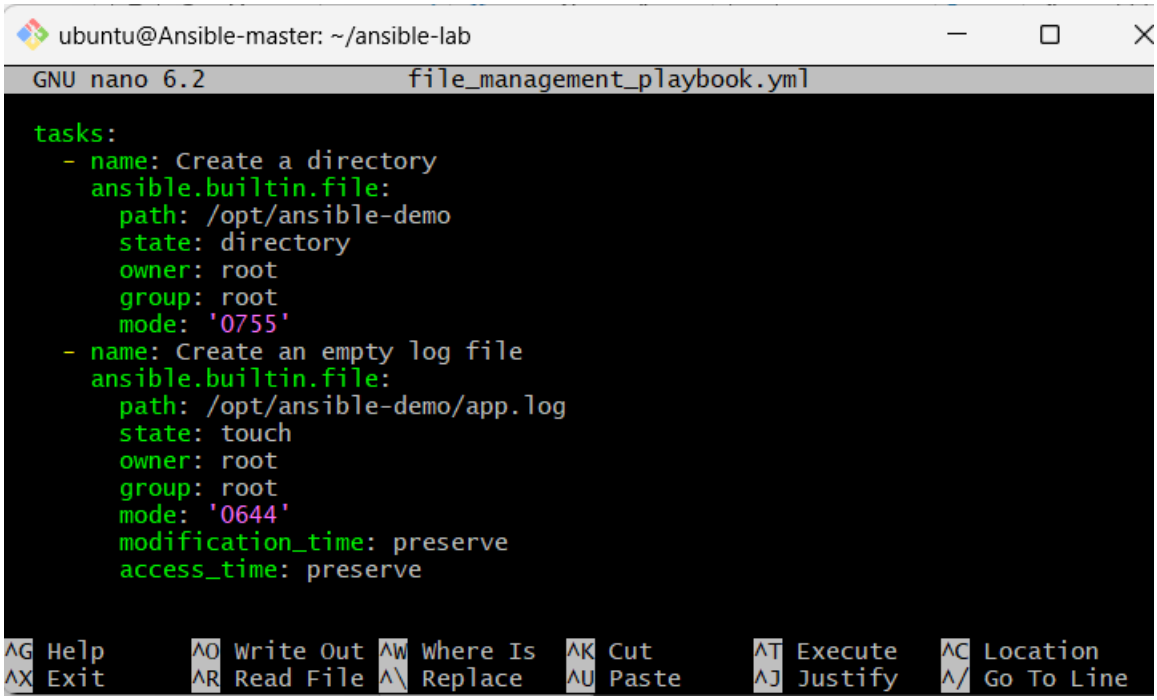
And it did!!

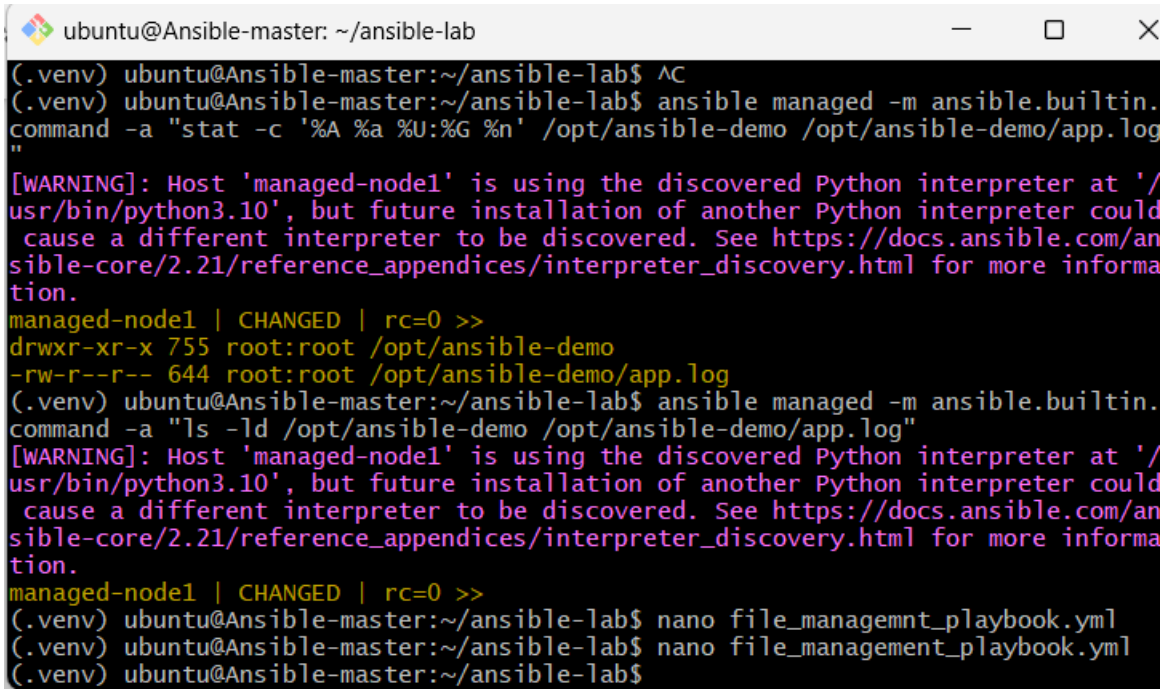


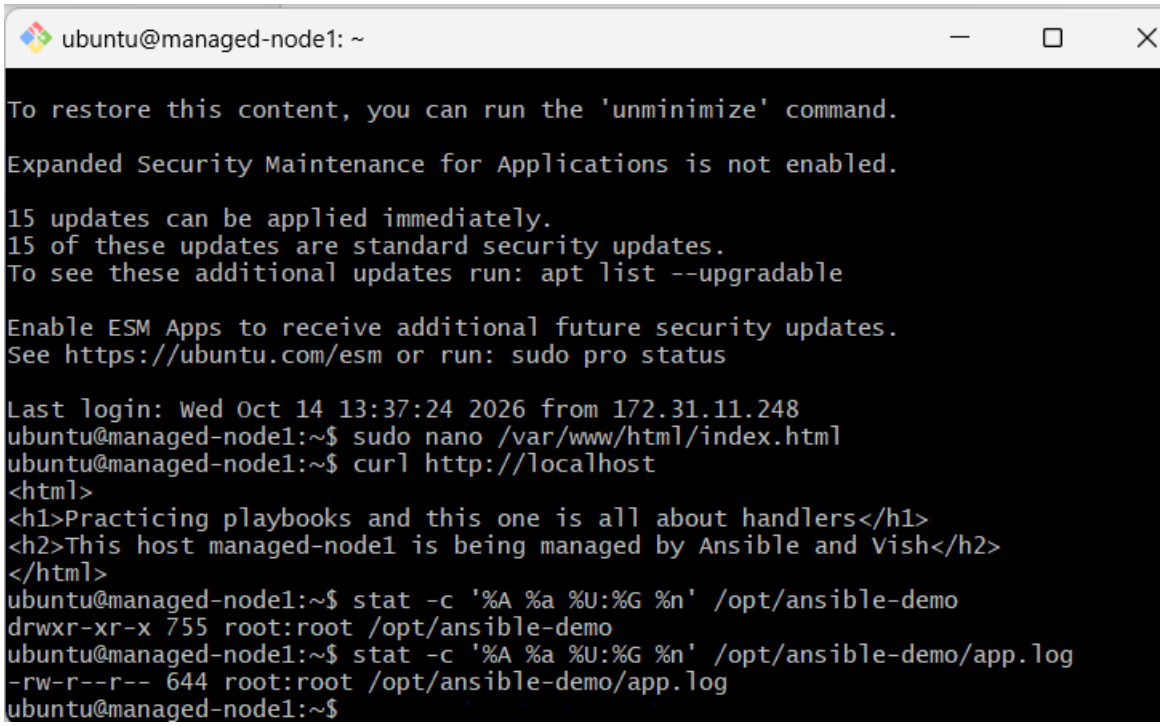
Practicing playbooks and this one is all about handlers

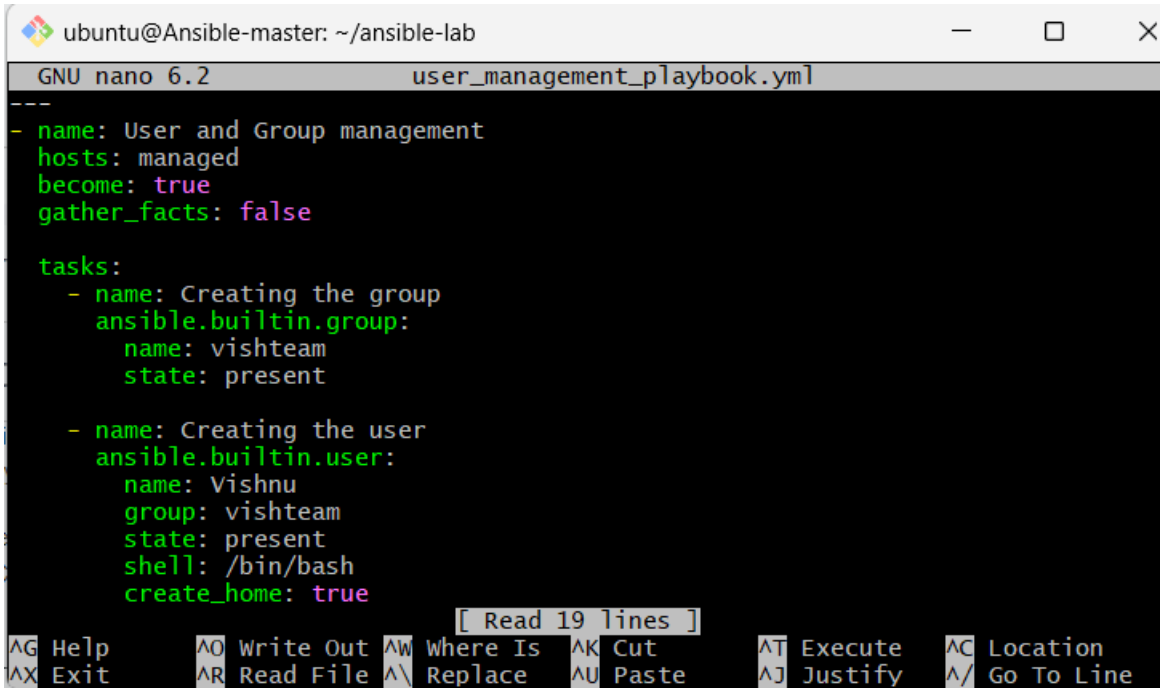
This host managed-node1 is being managed by Ansible and Vish



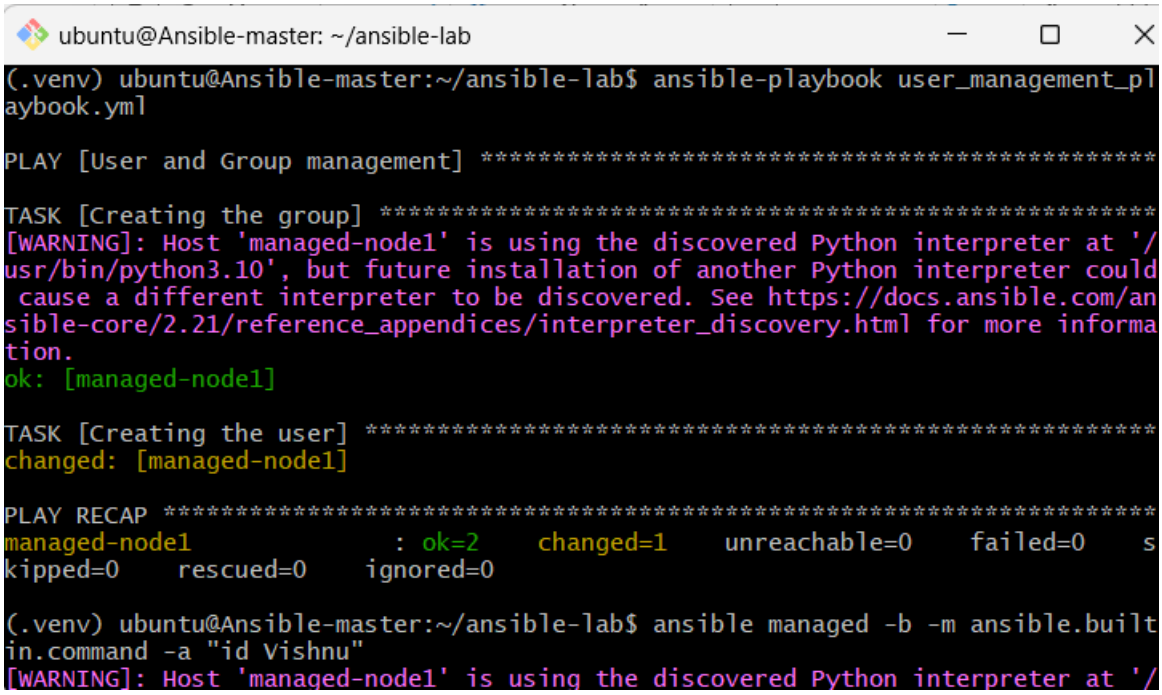

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 file_management_playbook.yml
tasks:
- name: Create a directory
ansible.builtin.file:
path: /opt/ansible-demo
state: directory
owner: root
group: root
mode: '0755'
- name: Create an empty log file
ansible.builtin.file:
path: /opt/ansible-demo/app.log
state: touch
owner: root
group: root
mode: '0644'
modification_time: preserve
access_time: preserve
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

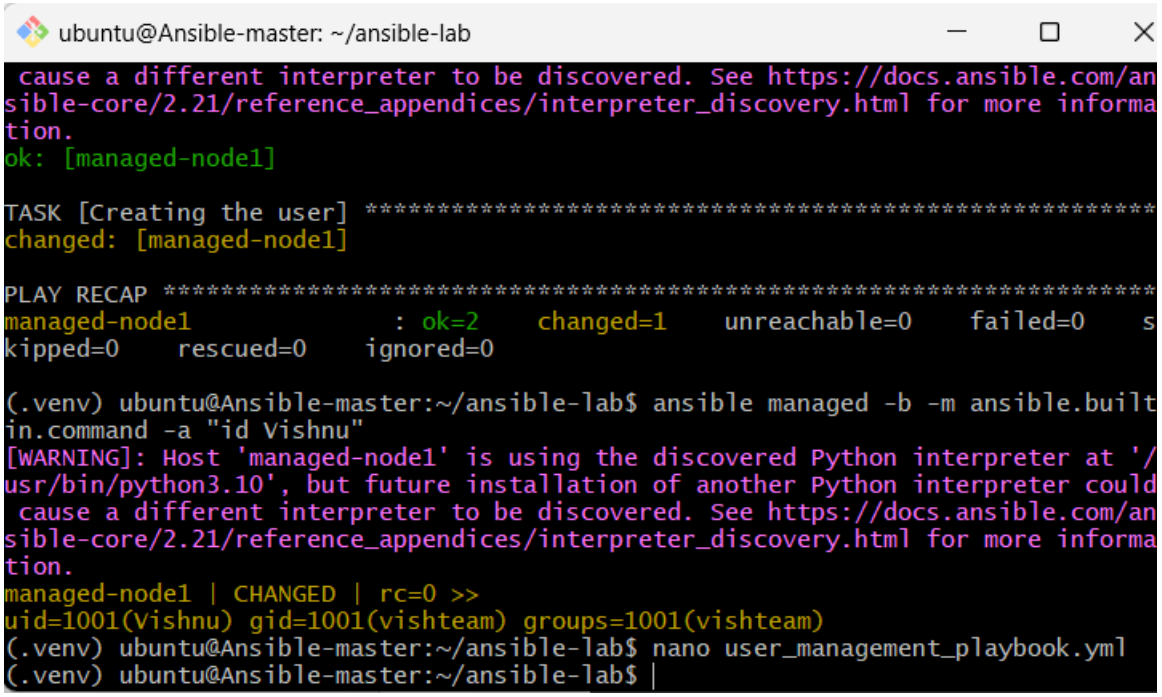

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ^C
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.command -a "stat -c '%A %a %U:%G %n' /opt/ansible-demo /opt/ansible-demo/app.log"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
drwxr-xr-x 755 root:root /opt/ansible-demo
-rw-r--r-- 644 root:root /opt/ansible-demo/app.log
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.command -a "ls -ld /opt/ansible-demo /opt/ansible-demo/app.log"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
(.venv) ubuntu@Ansible-master:~/ansible-lab$ nano file_managemnt_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ nano file_management_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$

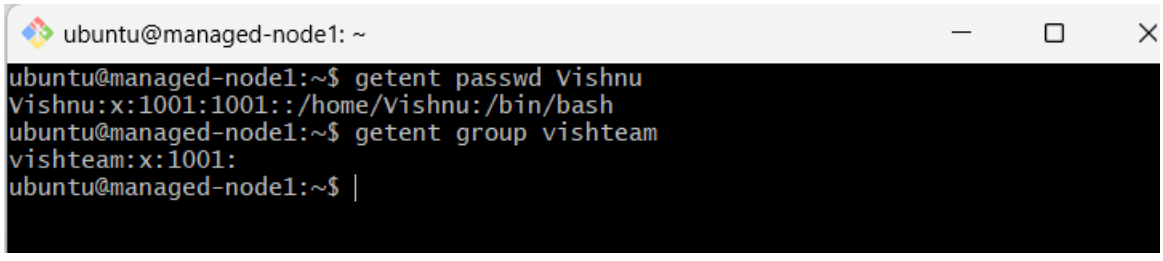

ubuntu@managed-node1: ~
To restore this content, you can run the 'unminimize' command.
Expanded Security Maintenance for Applications is not enabled.
15 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Wed Oct 14 13:37:24 2026 from 172.31.11.248
ubuntu@managed-node1:~$ sudo nano /var/www/html/index.html
ubuntu@managed-node1:~$ curl http://localhost
<html>
<h1>Practicing playbooks and this one is all about handlers</h1>
<h2>This host managed-node1 is being managed by Ansible and Vish</h2>
</html>
ubuntu@managed-node1:~$ stat -c '%A %a %U:%G %n' /opt/ansible-demo
drwxr-xr-x 755 root:root /opt/ansible-demo
ubuntu@managed-node1:~$ stat -c '%A %a %U:%G %n' /opt/ansible-demo/app.log
-rw-r--r-- 644 root:root /opt/ansible-demo/app.log
ubuntu@managed-node1:~$

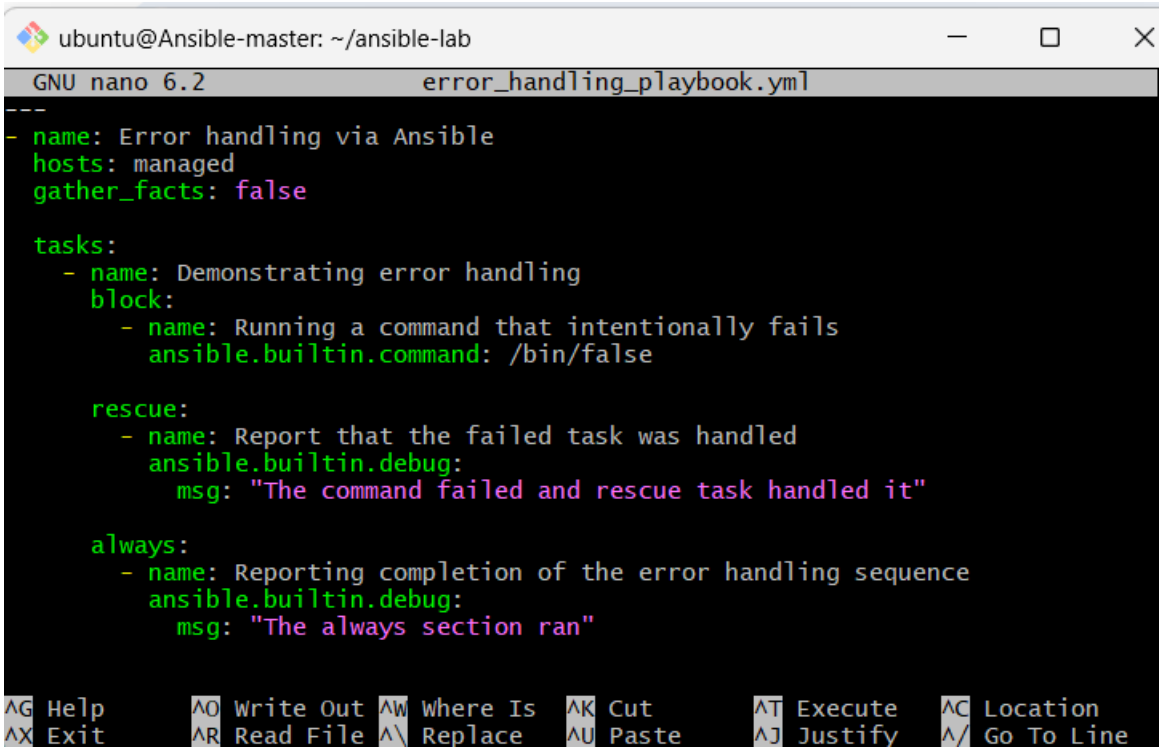

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 user_management_playbook.yml

- name: User and Group management
hosts: managed
become: true
gather_facts: false
tasks:
- name: Creating the group
ansible.builtin.group:
name: vishteam
state: present
- name: Creating the user
ansible.builtin.user:
name: Vishnu
group: vishteam
state: present
shell: /bin/bash
create_home: true
[Read 19 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook user_management_playbook.yml
PLAY [User and Group management] ***
TASK [Creating the group] **
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [managed-node1]
TASK [Creating the user] ***
changed: [managed-node1]
PLAY RECAP ***
managed-node1 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.builtin.command -a "id Vishnu"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/

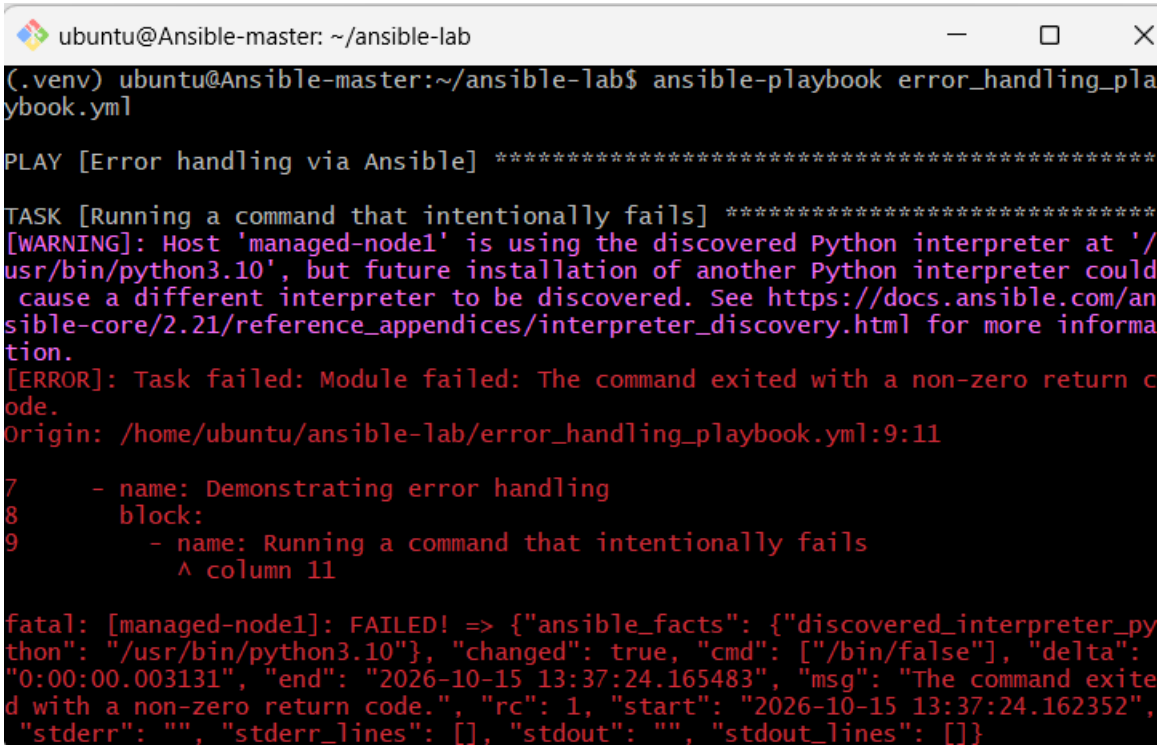

ubuntu@Ansible-master: ~/ansible-lab
cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [managed-node1]
TASK [Creating the user] ***
changed: [managed-node1]
PLAY RECAP ***
managed-node1 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.builtin.command -a "id Vishnu"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
uid=1001(Vishnu) gid=1001(vishteam) groups=1001(vishteam)
(.venv) ubuntu@Ansible-master:~/ansible-lab$ nano user_management_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$


ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ getent passwd Vishnu
Vishnu:x:1001:1001::/home/Vishnu:/bin/bash
ubuntu@managed-node1:~$ getent group vishteam
vishteam:x:1001:
ubuntu@managed-node1:~$



Ansible Error Handling Using Block, Rescue, and Always:

The /bin/false command intentionally fails with exit code 1 to demonstrate error handling. Ansible detects the failure and executes the rescue task to handle it, followed by the always task to report completion. The final recap shows rescued=1 and failed=0, confirming that the failure was successfully handled (Check the following outputs in the next page)



As expected, recued is set to 1 highlighting the sequence executed and rescue handled the error rather than the playbook ending with an unrecovered failure